CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

Learner name / date	Craig Temple 10/27/26
Import status	Load all 20 rows first

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	R03, R14
Q02	Work order is unique	2	10%	R05, R06
Q03	Status uses the agreed list	1	5%	R08
Q04	Duration is in range	2	10%	R10, R20
Q05	READY identifies an inspector	3	15%	R03, R07, R12

Distinct rows with any issue

Count each row once, even when several Q-rules flag it. Do not add the five counts.

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

R03 failed Q01 and Q05 because Machine_ID and Inspector_ID are both blank, despite its status being READY. The source is the inspection data, and this could cause issues with tracking the machine and assigning responsibility.

Next action: what to verify, who owns it, and how to retest

The inspection supervisor should verify the missing IDs against the original records. Once corrected, refresh Power Query and confirm both flags change from 1 to 0.

Limit: why you cannot guess a correction or prove accuracy

Passing the validation rules doesn't guarantee the data is accurate. We can confirm the records meet our requirements, but we can't verify the actual machine or inspection without comparing it against physical records.

Professional connection: one framework idea and how you used it

I used the validity concept from the government Data Quality Framework. Power Query allowed me to apply consistent validation rules and automatically flag records that failed.

Refresh test: before, changed value, after, and restored result

Before changes R01 Inspection_Minutes was set to '20' and Q04 flag was set to '1'. I updated R01Inspection_Minutes to '0', refreshed data, and the new Q04 flag was set to '0' indicating it was no longer within criteria of a whole number (not = null or 0). Passing

Export only Report: two landscape pages, page break before row 26. Review both pages.